An Execution-Time CBF-QP Safety Filter on Frozen MAPPO Formation Policies: A Pre-Specified Paired Evaluation in Simulation

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Abstract

We evaluate whether one execution-time, decentralised control-barrier-function quadratic-programme (CBF-QP) safety filter, placed between frozen multi-agent PPO (MAPPO) formation policies and a simulated team of three unicycle-modelled robots, meets non-inferiority criteria that the author fixed before the test. Nine frozen actors (three training runs, three checkpoints each), which had not met the study's filter-off competence gate (pooled over 2, 5 and 8 obstacles), were evaluated on reserved, previously unused layouts with one, two and three obstacles (9,000 paired trials per obstacle count) with the filter off, the filter on at $\alpha^\star = 10\,\mathrm{s}^{-1}$ (the largest gain the sampled-data condition admits, selected on validation layouts), and a brake applied on the same constraints. The filter reduced the share of trials ending in collision or QP infeasibility at every obstacle count (from 5.4% to 0.1%, 16.6% to 0.8% and 48.0% to 3.9%), while the share not reaching the goal rose (from 45.3% to 50.1%, 33.4% to 43.2% and 68.0% to 78.5%). Under the pre-set criteria it was *not shown useful* at any obstacle count: non-inferiority was not established for goal completion (margin 5 percentage points; at one obstacle the estimated loss, 4.81 points, was inside the margin but its upper bound was not) or for formation error over the common executed prefix (margin 0.01 m). The brake had no collisions, cannot be infeasible by construction, and reached the goal 1.0–3.2 points less often than the filter. Exploratory, post hoc analyses of the saved logs show that most lost completions ended as timeouts during which the filter was overriding the policy's command, and that every infeasible QP occurred with a constraint at its stand-off, where stopping is only marginally inadmissible. Claims cover this filter instantiation and these nine actors in one simulated scenario.

1 Introduction

Reinforcement-learning policies for multi-robot formations are commonly trained without safety constraints [1], and a control-barrier-function (CBF) safety filter [2] can be added at execution time without retraining. Whether such a filter is *useful*, here: reduces collision-or-infeasible outcomes without a loss of goal completion or formation quality larger than margins fixed in advance, is an empirical question for a given policy family. This paper reports a pre-specified paired evaluation of one such filter, a decentralised sampled-data CBF-QP, on frozen MAPPO [22] actors in a VMAS [23] differential-drive formation-navigation scenario, followed by clearly labelled exploratory analyses of where, among the paired trials, the lost completions occurred.

Contributions. Studies of safe multi-agent learning mostly add safety during training or learn it together with the policy. Among those that act at execution, one evaluates with and without its filter a policy that was trained with the filter, one compares the unfiltered policy mainly qualitatively with a centralised filter on double integrators, and one adds a learned correction to a

frozen policy without stating that its comparison is paired (Section 2, Table 1). This paper adds: (i) a paired evaluation, with criteria and margins fixed before the test and reserved test layouts, of a decentralised execution-time CBF-QP filter on a family of nine frozen MAPPO actors that never saw the filter, together with a brake comparator on the same layouts; (ii) a confirmatory result for these actors: collision-or-infeasible outcomes fell at every density, but non-inferiority in goal completion and formation error was not established, with outcomes, failure types and per-seed variation reported in full; (iii) exploratory, clearly labelled analyses of where the lost completions occurred (timeouts during which the filter overrode the policy's command, on layouts where the unfiltered runs passed close to obstacles) and of the infeasible states (each with a constraint at its stand-off); and (iv) the code, saved per-step logs of every final-test trial and scripts that regenerate every number, so the analyses can be repeated without re-running trials.

Research question. Does this execution-time CBF-QP filter, applied to frozen MAPPO formation policies trained without it, reduce collisions, and how does it change formation error, goal completion and sustained nonprogress? Each of these could worsen, stay the same or improve; the non-inferiority criteria of Section 5 would accept preservation or improvement.

Motivating hypotheses. The study was motivated by three hypotheses stated in the study's earlier research proposal, quoted here: "H1: collisions fall. H2: formation error and completion time rise, more so in denser layouts. H3: most failures under filtering come from deadlock or infeasible steps rather than slower progress." The proposal defined deadlock as "centroid progress below a set distance over a set window". The protocol implements this as a flag (named `stalled` in the saved files) set on a TIMEOUT whose centroid-to-goal distance fell by less than 0.05 m over the last 50 executed periods. Because the flag measures progress toward the goal, not distance travelled or stationarity, and a team moving sideways, circling, moving away or rearranging its formation also sets it, we call it *sustained nonprogress* and reserve "deadlock" for the quoted hypothesis. The expectation behind H3 had three grounds: CBF-QP filters can introduce undesirable stable equilibria [24, 28]; with the margin $\delta > 0$ used here, stopping is inadmissible just below the stand-off defined in Section 3; and the locked protocol predicted analytically that small gains would conflict with holding the formation (Section 8). Only targeted comparisons that vary these design elements could separate these causes; the analyses of Section 7 can suggest explanations but do not. These hypotheses motivate the design; they do not extend the locked confirmatory claims, which are the three criteria of Section 5. Section 8 maps each hypothesis to the evidence and states which evidence is confirmatory and which is exploratory.

2 Related work

Safe learning in robotics is surveyed in [1] and control barrier functions in [2]. Multi-agent approaches differ in when safety enters. Some apply a CBF shield during training [6] or learn barrier certificates together with the policy [7, 10, 8]; some filter during both training and execution [9, 5]; one learns a safety correction on top of a frozen MAPPO policy [4]; and one filters at execution a separately obtained RL planner, using a centralised QP for double-integrator agents [12]. Table 1 summarises them. Here *paired on/off* means that the same frozen policy is evaluated with and without the filter on matched layouts; a comparison between separately trained policies (a training ablation) does not count as paired in this sense. [5] does report a paired test in this sense, but for a policy trained with its filter, which is then deactivated; the present study evaluates policies that never saw the filter. The classifications were checked against the papers on 26 October 2026:

Table 1: Related multi-agent studies. "Paired on/off" = the same frozen policy evaluated with and without the filter on matched layouts. Checked against the full text (F) or the abstract only (A) on 26 October 2026.

Study	When safety enters	Paired on/off	Outcomes reported	Checked
Cai et al. [6]	CBF shield during training (also claimed for deployment)	No: separately trained agents	collisions, training reward	F
Qin et al. [7]	certificates learned with the policy; test-time refinement	not reported in abstract	safety, task completion	A
GCBF+ [10]	graph CBF and distributed policy learned together	not reported in abstract	safety, goal reaching	A
DGPPO [8]	graph CBF learned simultaneously with the policy	not reported in abstract	task performance, safety rate	A
Choi et al. [9]	CBVF filter in the second half of training and at execution	No: separately trained variants	goal reach, near collisions, conflicts, travel time	F
Deng et al. [4]	learned correction added to a frozen MAPPO policy	base policy reported; pairing not stated	return, arrival rate, violation rate	F
Dawood et al. [5]	MPC filter in training and execution	Yes, for a policy trained with the filter	goals, timeouts, collisions, time, formation error	F
Vinod et al. [12]	centralised QP filter at execution on a separately obtained RL planner (filter use in training not stated)	same policy and start states, reported mainly qualitatively	completion, deadlock, loitering	F
This study	decentralised CBF-QP at execution; policies trained without it	Yes: 9,000 matched layouts per density	collisions, infeasibility, completion, formation, nonprogress	–

against the full text for [6, 9, 5, 4, 12] and against the abstract only for [7, 10, 8], where "not reported in abstract" marks what could not be checked.

Sampled-data CBF formulations that account for inter-sample behaviour are studied in [13, 14, 15, 16]; the margin used here is of the input-independent kind, a bound over the whole input set that does not depend on the applied command. CBF-QP filters can introduce undesirable stable equilibria [24]. Decentralised equal-split ("half-responsibility") constraints go back to [3, 17], and stopping as a feasibility-preserving backup is used in [18]. The offset (barrier) point for unicycles is used in [19]. Hidden assumptions in CBF safety claims are discussed in [20]. Input-dependent inter-sample bounds used for trigger timing appear in [21]. Interval estimates and per-seed reporting for RL evaluation are argued for in [11].

Outside the multi-agent setting, execution-time correction of a learned policy has two main forms: shields that override unsafe actions of a reinforcement-learning agent against a formal specification [25], and CBF-based safety layers that minimally modify the policy's action, used during learning in [26]; [27] review safety filters as a common framework. These works mainly ask whether the filter keeps the system safe and how it affects learning; the cost of adding a filter at execution to a policy that never saw it, measured on matched trials with criteria fixed in advance, is the question studied here. For multi-robot CBF controllers, [28] show that symmetric configurations produce deadlocks, characterise them with KKT conditions and propose a decentralised resolution, and [24] show that CBF-QPs can introduce undesirable asymptotically stable equilibria. The present filter has no such resolution step, so its nonprogress behaviour is part of what is measured.

3 System and filter

Robots and task. Three differential-drive robots, modelled as unicycles with a box constraint on (v, ω) (wheel-speed coupling is not modelled), with axle-centre pose (x, y, θ), body disc of radius $R = 0.10\,\mathrm{m}$, and command $u = (v, \omega)$ in the box $\mathcal{U} = \{-0.5 \le v \le 0.5\ \mathrm{m/s},\ |\omega| \le 1\ \mathrm{rad/s}\}$ follow the unicycle model

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \omega, \tag{1}$$

with commands held over each control period $\Delta = 0.1\,\mathrm{s}$ (zero-order hold). The team moves a triangular formation (slot radius 0.30 m) through a field of K circular obstacles of radius $R_o = 0.20\,\mathrm{m}$ to a goal within a horizon of 400 periods. A *layout* fixes the obstacle centres and the initial poses: the team starts on its triangle slots around the centroid $(-1.5, 0)\,\mathrm{m}$ (position noise 0.03 m, heading noise 0.15 rad) and must bring its centroid to $(1.5, 0)\,\mathrm{m}$; the K obstacle centres are drawn uniformly in $[-0.8, 0.8] \times [-0.9, 0.9]\,\mathrm{m}$ with surface gaps of at least 0.10 m and clear of the start and goal; there are no walls. The *density* K is the number of obstacles in this fixed field. GOAL is reached at the first post-step state with the centroid within 0.10 m of the goal and the formation error $e(t)$ of Section 5 at most 0.10 m; COLLISION (robot–obstacle or robot–robot overlap) is scored on the swept path and ends the trial; an infeasible QP ends the trial as INFEASIBLE at that period, before the period's motion; otherwise the trial ends as TIMEOUT.

Offset point and barriers. The filter constrains the point $p = (x + \ell\cos\theta,\ y + \ell\sin\theta)$, $\ell = 0.05\,\mathrm{m}$, for which

$$\dot{p} = J(\theta)\,u, \qquad J(\theta) = \begin{bmatrix} \cos\theta & -\ell\sin\theta \\ \sin\theta & \ell\cos\theta \end{bmatrix}, \qquad \det J = \ell, \tag{2}$$

so every barrier below has relative degree one in (v, ω). For an obstacle with centre c and robots $i \ne j$ the barriers are

$$h_o = \|p - c\|^2 - D_o^2, \quad D_o = R + \ell + R_o; \qquad h_{ij} = \|p_i - p_j\|^2 - D_{rr}^2, \quad D_{rr} = 2(R + \ell). \tag{3}$$

Because $\|x - p\| = \ell$, the body disc lies inside the *cover disc* of radius $R + \ell$ around p, so $h \ge 0$ implies that the body discs do not overlap ($h > 0$: strictly separated), for every heading. The converse does not hold: heading straight at an obstacle, a robot has $h_o < 0$ at any body clearance below $2\ell = 10\,\mathrm{cm}$ (Fig. 1).

Sampled-data condition. The implemented filter uses the following lemma with a two-sided bound on $\ddot{h}$; only the lower bound is needed.

Lemma 1 (one-sided form). Let u be held on $[t_k, t_k + \Delta]$, $\ddot{h} \ge -M$ on that interval with $M \ge 0$, $\alpha\Delta \le 1$ and $h(t_k) \ge 0$. If $\dot{h}(t_k) + \alpha h(t_k) \ge \delta = M\Delta/2$, then $h(t_k + s) \ge (1 - \alpha s)\,h(t_k) \ge 0$ for all $s \in [0, \Delta]$.

This is the second-order approach of [14] (Theorem 3, stated there with the opposite sign convention for h; $\gamma \le 1$ corresponds to $\alpha\Delta \le 1$). With $\bar{v} = \max|v| = 0.5\,\mathrm{m/s}$, $\bar{\omega} = \max|\omega| = 1\,\mathrm{rad/s}$, $V_p = \sqrt{\bar{v}^2 + \ell^2\bar{\omega}^2} \approx 0.5025\,\mathrm{m/s}$ (an upper bound on $\|\dot{p}\|$) and $A_p = \bar{\omega}V_p$ (an upper bound on $\|\ddot{p}\|$ under a held command), the implemented margins use the two-sided bounds of the design document,

$$\delta_o(d) = \tfrac{\Delta}{2}\big[2V_p^2 + 2(d + V_p\Delta)A_p\big], \qquad \delta_{ij}(d) = \tfrac{\Delta}{2}\big[8V_p^2 + 4(d + 2V_p\Delta)A_p\big], \tag{4}$$

where d is the barrier-point distance at t_k: $d = \|p - c\|$ for an obstacle row and $d = \|p_i - p_j\|$ for a pair row. These bounds follow from $\|\ddot{p}\| \le A_p$ under a held command, $\ddot{h}_o = 2\|\dot{p}\|^2 + 2(p - c)^\top\ddot{p}$ and

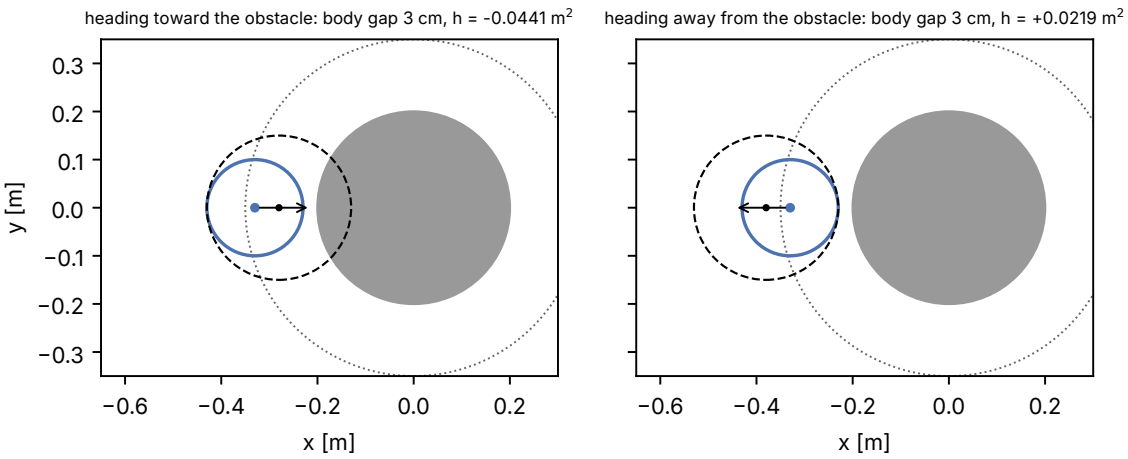


Figure 1: Analytic schematic (no data). Body disc (solid), axle centre, offset point p and cover disc of radius $R+\ell$ (dashed) next to an obstacle (grey) whose barrier boundary $\|p-c\| = D_o$ is dotted. With the same 3 cm body clearance, the cover barrier is negative when the robot heads toward the obstacle and positive when it heads away.

$\|p(t)-c\| \le d + V_p\Delta$ within the period (pair rows: relative speed and acceleration at most $2V_p$ and $2A_p$). Only the lower bound $\ddot{h} \ge -M$ is needed, so the $2V_p^2$ and $8V_p^2$ terms add conservatism. At $\alpha\Delta = 1$ the lemma still gives $h \ge 0$ over the whole period, but no strictly positive endpoint bound.

Equal split and the per-robot QP. With $g_o = 2J(\theta_i)^\top(p_i - c)$ and $g_{ij} = 2J(\theta_i)^\top(p_i - p_j)$, robot i solves at every period

$$\begin{aligned} &\min_{u\in\mathcal{U}} \tfrac{1}{2}\|u - u_{\text{nom}}\|^2 \\ &\text{s.t. } g_o^\top u \;\ge -\alpha h_o + \delta_o(d) \qquad \text{obstacles with } d \le 1.0\,\text{m}, \\ &\qquad g_{ij}^\top u \;\ge -\tfrac{\alpha}{2} h_{ij} + \tfrac{1}{2}\delta_{ij}(d) \quad \text{neighbours with } d \le 1.5\,\text{m}, \end{aligned} \tag{5}$$

where u_{nom} is the frozen actor's deterministic command. The sensing ranges exceed $D_o + V_p\Delta$ and $D_{rr} + 2V_p\Delta$ (both about 0.40 m), so an omitted barrier cannot reach zero within the period. An object outside range does not disappear from the safety assessment: rows are rebuilt at every period, and a neighbour beyond 1.5 m must close 1.20 m of barrier-point distance to reach D_{rr} at no more than $2V_p\Delta \approx 0.10$ m per period (at least 12 periods), and an obstacle beyond 1.0 m must close 0.65 m at no more than $V_p\Delta \approx 0.05$ m per period (at least 13 periods), so it re-enters the constraint set long before it could matter. This relies on A2–A3 below: within range the filter uses exact obstacle centres and exact, undelayed neighbour poses. In the final test, 1,664 of 21,074,097 robot-pair periods (0.008%) had a pair beyond 1.5 m (largest 1.78 m), so limited communication was in effect not tested. Each robot takes half of every pair condition (equal split); the two halves sum to the pair condition of Lemma 1 when both robots compute their rows from the same state, apply their commands over the same period and both QPs are feasible. In this simulation the first two hold by construction (every robot reads the exact, synchronous joint state), so "decentralised" refers to how the computation is split; with inconsistent or delayed states no guarantee is claimed.

Stand-off. Because $\delta > 0$, the zero command satisfies an obstacle row only if $h_o \ge h^\star := \delta_o(d)/\alpha$, and a pair row only if $h_{ij} \ge \delta_{ij}(d)/\alpha$. We call $h^\star$ the *stand-off*: the smallest barrier value at which a robot may stop. At $\alpha^\star$ and $d = D_o$ the obstacle stand-off lies 0.64 cm of barrier-point distance

beyond D_o, so a stopped robot needs a body clearance from an obstacle between about 0.6 cm (heading away) and 10.6 cm (heading toward it).

Guarantee and assumptions. By induction over periods, Lemma 1 and the equal split keep every barrier non-negative for all t, so the bodies never overlap, under these assumptions: (A1) the realised motion is the exact unicycle arc of the held command; (A2) exact, noise-free states and obstacle positions; (A3) synchronous, undelayed neighbour states within range; (A4) zero-order hold over Δ, with the QP solved within the period; (A5) $h \geq 0$ for every barrier at the start; (A6) static circular obstacles and no walls; (A7) the command box contains $u = 0$ and is shared by the policy clip and the QP; (A8) $\alpha\Delta \leq 1$, $\ell > 0$ and the range rule above; (A9) every QP is feasible. A trial whose QP becomes infeasible violates A9 and ends as INFEASIBLE. In simulation A1 holds to the integrator's float32 accuracy. The two-variable QP is solved by enumerating its candidate optima (the unconstrained optimum, the projection onto each constraint line and every pairwise vertex) and keeping the closest feasible one; this is implemented in floating-point arithmetic with a feasibility tolerance of 10^{-9} on normalised rows; the QP is INFEASIBLE when no candidate satisfies every enabled row within this tolerance. There is no fallback, so an infeasible QP ends the trial. A period is an *intervention* when $\|u - \mathrm{clip}(u_{\mathrm{nom}})\|_\infty > 10^{-6}$. The B-match comparator (a brake matched to the filter's rows) applies $(0, \bar{\omega}_{\mathrm{nom}})$, with $\bar{\omega}_{\mathrm{nom}}$ the clipped nominal turn rate, whenever the clipped nominal command violates one of these rows at the same gain. It solves no QP, so it cannot be INFEASIBLE by construction. With $v = 0$ the body disc does not move, so a braked robot cannot strike a static obstacle during that period, but B-match carries no barrier-level guarantee (p rotates, so h can decrease) and none against a moving neighbour.

4 Policy and training

Observation and reward. Each robot observes, in its body frame: its slot error relative to the centroid, the centroid-to-goal vector (norm-clipped at 2.0 m), teammates' relative positions and headings, the eight nearest obstacles within 2.0 m (relative centre, surface gap, present flag) and an agent one-hot. The per-period reward is $10\cdot$(centroid progress toward the goal) $-1\cdot$(mean slot error) $-0.01\cdot$(control effort) $-10\cdot$[collision] $-0.2\cdot\sum_{\mathrm{pairs}}\max(0, 1 - \mathrm{clearance}/0.15\,\mathrm{m})$, summed over robot–obstacle and robot–robot pairs (surface clearance); control effort is the mean over robots of $(v/0.5)^2 + \omega^2$; a collision ends the episode and a timeout is bootstrapped. The filter is never used in training.

MAPPO and curriculum. A compact PyTorch implementation of MAPPO [22] (shared Gaussian actor with tanh-bounded mean, centralised critic, GAE with $\gamma = 0.99$, $\lambda = 0.95$, clip 0.2, 10 epochs, 16 minibatches, learning rates 3×10^{-4} actor and 10^{-3} critic, no entropy bonus, gradient-norm clip 0.5, hidden layers 128×128 (actor) and 256×256 (critic), initial action standard deviation half of each half-range, 384 parallel environments of 64 steps) was trained with three seeds through a curriculum $K = 0 \to 1 \to 2 \to 3 \to (2, 5, 8)$ with maximum stage budgets of 1, 2, 4, 4 and 8 million environment steps; a stage ended early once the GOAL rate on the validation layouts (Section 5) reached 80% at every stage density, which happened in the $K = 0$ and $K = 1$ stages for all three seeds. The final stage trains on $K = 2, 5, 8$ only; $K = 1$ and $K = 3$ are absent from it but were encountered in earlier stages. The implementation was sanity-checked only on the stock VMAS navigation task, where the deterministic policy reached 99.0% success on 200 fixed episodes (uniform random: 0.0%).

Table 2: Filter-off baseline on validation layouts (200 per cell): GOAL / COLLISION / TIMEOUT (%) per actor and obstacle count K (descriptive context; $K = 4$ was an intermediate diagnostic).

actor	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$	$K = 8$
s0_early	82/5/13	52/26/22	16/63/22	6/84/10	0/98/2	0/100/0
s0_middle	64/3/33	48/24/28	14/58/28	7/80/12	0/98/2	0/100/0
s0_late	56/4/39	50/24/26	16/58/27	5/80/15	0/98/2	0/100/0
s1_early	17/12/72	56/21/23	25/56/20	12/84/4	0/97/2	0/100/0
s1_middle	42/10/48	75/13/12	35/48/17	18/76/6	1/93/6	0/100/0
s1_late	44/5/50	78/10/12	25/48/26	20/74/6	4/90/6	0/100/0
s2_early	24/4/72	72/11/16	58/28/14	19/70/11	2/96/2	0/100/0
s2_middle	53/5/42	75/14/11	50/31/19	5/82/12	0/96/3	0/100/0
s2_late	76/5/18	80/16/5	49/34/18	18/78/4	0/98/2	0/100/0

Checkpoints and gate history. The nine actors are the early, middle and late checkpoints of the final curriculum stage of each seed, fixed by an iteration rule (first saved checkpoint with $r \geq \lceil n/3 \rceil$, $r \geq \lceil 2n/3 \rceil$, and $r = n$, where r is the iteration counted within the final stage and $n = 326$ is its length). The three checkpoints of one seed come from the same training run and are not independent policies. Seed 1 crashed at global iteration 565 (a layout-sampler hang) and was resumed from the state it had saved at iteration 546: actor, critic, optimiser states, value normaliser and action-sampling generator were restored bit for bit, but the environments, the position in the training-layout chunk and the global random-number state were not, and iterations 547–565 (466,944 environment steps) were discarded. Actors s1_middle and s1_late come from the resumed run; s1_early predates the crash The global random-number state is used neither for action sampling nor for layouts. The resumed actors cannot be regenerated by re-running seed 1; the saved checkpoint files are the reference. The filter-off competence gate fixed before training (pooled GOAL $\geq 50\%$ over $K = 2, 5, 8$; below it the policy "rarely reaches the goal") was not met by any checkpoint: the best pooled GOAL rate was 11.0% for a first configuration (four observed obstacles, no proximity penalty, no curriculum) and 28.7% after the observation, reward, PPO settings and curriculum were revised (the configuration described above). $K = 5, 8$ were therefore moved to stress cells and $K = 1, 2, 3$ became the primary densities, chosen after the actors' filter-off validation results had been seen so that both completions and collisions occur. The results therefore concern a weak, collision-prone policy family; they do not predict the filter's cost on policies that keep more clearance. Table 2 shows the filter-off baseline of the nine actors on validation layouts.

5 Experimental design

The protocol was locked by the author before any filter-on data with the frozen MAPPO actors existed; the margins and decisions were chosen without advisor review (Section 12). Filter-on results did exist before the lock in two other forms: engineering stress tests of the filter with random and adversarial stand-in controllers (4,200 paired trials) and paired demonstration runs with an obstacle-blind stand-in controller. Neither used the frozen MAPPO actors. The frozen-MAPPO development sweep on validation layouts followed the lock, and the reserved final evaluation followed the gain selection. *Validation layouts* (200 per K, seed $2{,}000{,}000 + K$) were used to advance curriculum stages, for the filter-off gate and Table 2, and for gain selection. *Final-test layouts* (1,000 per actor and K) were not used before the final test.

The gain was selected on existing validation layouts from $\alpha \in \{0.5, 1, 2, 5, 10\}\,\mathrm{s}^{-1}$ by the highest

pooled filter-on GOAL count over the nine actors and $K = 1, 2, 3$ (ties: fewer INFEASIBLE, then the larger gain). The final test used 1,000 reserved layouts per actor and density, each run with the filter off, the filter on at $\alpha^\star$ and B-match at $\alpha^\star$. Usefulness at a density required all of: (a) a one-sided exact McNemar reduction in COLLISION-or-INFEASIBLE; (b) GOAL loss (off minus on) with one-sided 98.33% upper bound at most 0.05; (c) the equal-actor-weighted mean increase in formation error over the common executed prefix with stratified bootstrap 98.33% upper bound at most 0.01 m. Within a density the three criteria form an intersection–union test, so no split across (a)–(c) is needed; each is tested one-sided at 0.05/3, a Bonferroni split over the three density claims, so the familywise error over the three usefulness claims is at most 0.05. Criterion (a) is decided by the exact McNemar test (its interval bound is reported); (b) is decided by the interval bound. For (a) and (b) the paired difference in proportions has a two-sided 96.67% Newcombe score interval [29] (Wilson score limits combined through the estimated correlation $\hat{\phi}$), whose appropriate limit gives the one-sided 98.33% bound. The locked protocol labels this interval "method 10" but describes it as combining the Wilson limits through $\hat{\phi}$ "without continuity correction", which is Newcombe's method 8; the analysis code, committed before any frozen-MAPPO filter-on data, implements that description. Method 10 applies a continuity correction to $\hat{\phi}$. The label inconsistency was found after the final test. Recomputing every bound for (a) and (b) with method 10 changes it by less than 0.001 percentage points and changes no criterion; the reported numbers are those of the locked code. The margins are the author's judgement, fixed without advisor review or external calibration; for any other margin, the 98.33% bounds in Table 3 give the smallest margin each criterion would have met. Per-seed and per-actor results, episode length, reached-goal-region (centroid within 0.30 m of the goal before any collision or infeasibility), final centroid-to-goal distance, TIMEOUT and the sustained-nonprogress flag were pre-declared as secondary and descriptive.

Formation endpoint (criterion (c)). For a trial, let T be the number of control periods actually executed: a period that ends in COLLISION counts, while an attempted period that an infeasible QP prevented does not. Let $x_r(t)$ be robot r's position at the post-step state after period t, $\bar{x}(t)$ the team centroid and s_r its slot offset; the formation error is

$$e(t) = \tfrac{1}{3} \textstyle\sum_{r=1}^{3} \big\| x_r(t) - \bar{x}(t) - s_r \big\|, \qquad t = 1, \ldots, T, \tag{6}$$

so the initial state and the frozen samples after termination are excluded. For pair i with executed counts T_{off} and T_{on}, the common executed prefix has length $L_i = \min(T_{\text{off}}, T_{\text{on}})$ and, for $L_i > 0$,

$$\Delta e_i = \frac{1}{L_i} \sum_{t=1}^{L_i} \big[e_{\text{on}}(t) - e_{\text{off}}(t) \big]. \tag{7}$$

For pairs with $L_i = 0$ (an arm infeasible at its first period) the formation difference is undefined; they are excluded from (c) and never counted as zero. Each arm keeps its recorded outcome in (a) and (b): an infeasible arm counts as a failure and as not reaching GOAL, the other arm keeps its own outcome, and no pair is assigned a GOAL loss by the exclusion. With $\mathcal{E}_p$ the eligible pairs of actor p, the estimand is the equal-actor mean

$$\overline{\Delta e} = \frac{1}{9} \sum_{p=1}^{9} \frac{1}{|\mathcal{E}_p|} \sum_{i \in \mathcal{E}_p} \Delta e_i. \tag{8}$$

Its upper bound is a stratified paired percentile bootstrap: pairs are resampled with replacement within each actor, $\overline{\Delta e}$ is recomputed with the same equal weights, $B = 20{,}000$ replicates are drawn

Table 3: Confirmatory results, filter on ($\alpha^\star = 10$) versus filter off, nine actors pooled, 9,000 pairs per density. Differences in percentage points (pp); one-sided bounds at 98.33%; 95% intervals descriptive.

	$K = 1$	$K = 2$	$K = 3$
COLLISION-or-INFEASIBLE, off → on (%)	5.4 → 0.1	16.6 → 0.8	48.0 → 3.9
(a) difference on−off [pp]	−5.30	−15.72	−44.08
98.33% upper; 95% CI	−4.82; [−5.78, −4.85]	−14.92; [−16.49, −14.98]	−42.95; [−45.11, −43.04]
one-sided exact p; criterion	2.6×10^{-144}; met	$< 10^{-300}$; met	$< 10^{-300}$; met
GOAL off / on (%)	54.7 / 49.9	66.6 / 56.8	32.0 / 21.5
(b) GOAL loss off−on [pp]	4.81	9.81	10.56
98.33% upper (margin 5); criterion	5.31; not met	10.59; not met	11.33; not met
(c) formation increase [m]	0.0117	0.0202	0.0278
bootstrap 98.33% upper (margin 0.01)	0.0127	0.0213	0.0291
normal cross-check; eligible / excluded	0.0126; 9,000/ 0	0.0213; 9,000/ 0	0.0291; 9,000/ 0
criterion	not met	not met	not met
Verdict	not shown useful	not shown useful	not shown useful

with `numpy.random.default_rng(20261026)`, and the bound is the $1 - 0.05/3$ quantile of the replicates (`numpy.quantile`, method `linear`). A normal bound from the per-actor variances is reported as a cross-check. If any actor has fewer than 2 eligible pairs, criterion (c) is inconclusive and the weights are not renormalised. In the final test every pair was eligible (Table 3). The common prefix had median length 87, 73 and 67 periods at $K = 1, 2, 3$ (quartiles 71–400, 66–86 and 32–101), so (c) mainly reflects early transit; when the unfiltered run collides, the prefix ends at that collision and excludes the filter's later avoidance manoeuvre.

6 Results

Gain selection (development data). Pooled filter-on GOAL counts over 5,400 validation trials were 0, 255, 1,184, 1,936 and 2,224 for $\alpha = 0.5, 1, 2, 5, 10$, with 1,251, 4, 11, 61 and 129 INFEASIBLE outcomes, so $\alpha^\star = 10$ (Fig. 2). GOAL rose with every step up the grid, and $\alpha^\star$ is also the largest gain with $\alpha\Delta \leq 1$: the evaluated filter is the least conservative admissible member of this family on the development data.

Confirmatory results. Table 3 gives the confirmatory results. Criterion (a) was met at every density; (b) and (c) were met at none, so the filter was *not shown useful under the pre-set criteria* at $K = 1, 2, 3$. At $K = 1$ the GOAL-loss estimate (4.81 pp) lies inside the 5 pp margin and only its upper bound (5.31 pp) exceeds it; at $K = 2, 3$ the estimates exceed the margin. Read together, the filter changed collision-or-infeasible outcomes by −5.30, −15.72 and −44.08 pp and lost 4.81, 9.81 and 10.56 pp of GOAL; how these trade off depends on the relative cost of a collision and a timeout in an application, and the margins encode one such judgement. With the filter on there were 0, 0 and 0 collisions and 6, 75 and 352 INFEASIBLE trials; TIMEOUT rose from 39.9%, 16.9% and 20.0% to 50.1%, 42.4% and 74.6% (Fig. 3). B-match had 0, 0 and 0 collisions and 0, 0 and 0 INFEASIBLE outcomes, and reached GOAL in 48.8%, 53.5% and 19.3% of trials; the filter reached GOAL 1.0, 3.2 and 2.2 percentage points more often (descriptive). B-match solves no QP, so its zero INFEASIBLE count holds by construction.

Filter on versus B-match (pre-declared secondary, descriptive). The locked protocol pre-declared a comparison of the filter with B-match at $\alpha^\star$ on the same pairs and endpoints, with

Table 4: Filter on versus B-match at $\alpha^\star$ (pre-declared secondary, descriptive; nothing confirmatory follows). Paired differences, filter on minus B-match, on all 9,000 pairs per density: percentage points with two-sided 95% Newcombe method-10 intervals; formation: equal-actor mean common-prefix difference in metres with a 95% stratified paired bootstrap interval. B-match cannot be INFEASIBLE by construction.

	$K = 1$	$K = 2$	$K = 3$
GOAL [pp]	1.03	3.23	2.18
95% interval	[0.81, 1.25]	[2.79, 3.67]	[1.79, 2.57]
COLLISION-or-INFEASIBLE [pp]	0.07	0.83	3.91
95% interval	[0.01, 0.15]	[0.66, 1.04]	[3.53, 4.33]
TIMEOUT [pp]	−1.10	−4.07	−6.09
95% interval	[−1.33, −0.87]	[−4.54, −3.59]	[−6.64, −5.54]
formation, common prefix [m]	−0.0014	−0.0030	0.0004
95% interval	[−0.0018, −0.0010]	[−0.0037, −0.0024]	[−0.0005, 0.0013]

intervals, and states that nothing confirmatory follows from it. Table 4 gives the paired differences, always filter on minus B-match, on all 9,000 pairs per density. The intervals are two-sided 95% Newcombe method 10 intervals: Wilson score limits for each arm, combined through the continuity-corrected correlation $\hat{\phi}$, whose numerator $ad-bc$ is replaced by $\max(ad-bc-n/2, 0)$ when $ad-bc > 0$ (a: event in both arms, b: filter on only, c: B-match only, d: neither). They were computed with a new implementation after the method 8/method 10 discrepancy described in Section 5 was found; the confirmatory intervals are unchanged. B-match had no COLLISION or INFEASIBLE outcome, so for that endpoint one marginal variance is zero, the correlation is undefined, and $\hat{\phi} = 0$ is used. The formation row is the equal-actor mean over pairs of $\frac{1}{L}\sum_{t=1}^{L}[e_{\text{on}}(t) - e_{\text{B-match}}(t)]$, with $L = \min(T_{\text{on}}, T_{\text{B-match}})$ and $e(t)$ recomputed from the saved poses, with a 95% stratified paired percentile bootstrap within actor (the locked settings: 20,000 resamples, seed 20261026); 0, 0 and 0 pairs had $L = 0$ and were excluded. The filter reached GOAL more often than B-match and timed out less often at every density, and had more COLLISION-or-INFEASIBLE outcomes; B-match cannot be infeasible by construction, so that row compares the filter's infeasible terminations with none. No test or verdict is attached to these differences.

Reached goal region (pre-declared secondary, descriptive). The share of trials whose centroid came within 0.30 m of the goal before any collision or infeasibility fell from 94.6% to 87.1%, 83.2% to 69.5% and 51.3% to 35.0% at $K = 1, 2, 3$ (paired differences −7.5 [−8.1, −7.0], −13.7 [−14.4, −12.9] and −16.3 [−17.2, −15.5] percentage points, 95% intervals). Which pairs lost a completion, and whether their filtered runs approached the goal, is counted per pair in Section 7 (exploratory). The remaining pre-declared secondary results (final centroid-to-goal distance, formation error over the full trajectory in pairs where both arms reached GOAL, intervention and brake rates, sampled barrier minima, and the remaining filter-on versus B-match comparisons) are reported in the repository.

Per seed and per actor (pre-declared secondary, descriptive). Table 5 gives the three locked endpoints per training seed with 95% intervals; no test or verdict is made per seed, and a pooled result does not establish usefulness, or its absence, for each seed or actor. Per-actor GOAL losses vary widely (Fig. 4). Heterogeneity is largest at $K = 1$: seeds 0 and 2 had GOAL losses of 2.9 and 4.0 pp and formation increases of 0.0047 and 0.0052 m, seed 1 had 7.5 pp and 0.0252 m, and 5 of the nine actors had descriptive 95% intervals inside both margins (none at $K = 2, 3$).

Table 5: Locked endpoints per training seed (pre-declared secondary, descriptive; three actors and 3,000 pairs per row; 95% intervals; no tests).

seed	K	COLLISION-or-INFEASIBLE off → on (%)	(a) difference [pp]	(b) GOAL loss [pp]	(c) formation increase [m]
0	1	3.9 → 0.0	−3.9 [−4.7, −3.3]	2.9 [2.3, 3.5]	0.0047 [0.0040, 0.0054]
1	1	8.5 → 0.2	−8.3 [−9.4, −7.4]	7.5 [6.5, 8.5]	0.0252 [0.0228, 0.0276]
2	1	3.7 → 0.0	−3.7 [−4.4, −3.0]	4.0 [3.3, 4.8]	0.0052 [0.0043, 0.0061]
0	2	24.5 → 1.3	−23.2 [−24.7, −21.7]	6.4 [5.3, 7.5]	0.0222 [0.0204, 0.0241]
1	2	14.0 → 0.5	−13.5 [−14.8, −12.3]	9.0 [7.8, 10.3]	0.0201 [0.0182, 0.0219]
2	2	11.1 → 0.7	−10.5 [−11.6, −9.4]	14.0 [12.6, 15.3]	0.0183 [0.0167, 0.0199]
0	3	62.4 → 7.0	−55.4 [−57.2, −53.6]	7.9 [6.9, 8.9]	0.0276 [0.0254, 0.0298]
1	3	51.1 → 2.5	−48.6 [−50.4, −46.8]	6.7 [5.6, 7.8]	0.0261 [0.0239, 0.0283]
2	3	30.5 → 2.3	−28.3 [−29.9, −26.7]	17.1 [15.5, 18.6]	0.0297 [0.0277, 0.0317]

Table 6: Time to goal in joint-success pairs (periods of 0.1 s; test arm minus filter off; derived from the pre-declared episode-length data; exploratory, post hoc joint-success analysis; Wilcoxon p with zero differences discarded, not inferential).

arm	K	pairs	fewest per actor	equal-actor mean	median	HL	slower (%)	Wilcoxon p
filter on	1	4,472	199	0.30	0.0	0.0	9.0	2.6×10^{-57}
filter on	2	4,968	406	0.68	0.0	0.0	23.9	1.9×10^{-89}
filter on	3	1,815	65	2.09	0.0	0.5	41.3	5.1×10^{-90}
B-match	1	4,385	199	0.47	0.0	0.0	10.1	1.3×10^{-59}
B-match	2	4,671	370	−0.24	0.0	0.0	27.3	5.5×10^{-92}
B-match	3	1,629	53	2.62	0.0	1.0	44.3	3.6×10^{-90}

The pooled $K = 1$ result therefore depends strongly on seed 1. Two of its actors come from the resumed run, but its pre-crash early actor has the largest formation increase of all nine, so these data do not single out the resume. All intervals reflect layout sampling for these fixed actors, not variation across training runs, of which there are three.

Time to goal (derived from the pre-declared episode-length data; exploratory, post hoc joint-success analysis). In pairs where both arms reached GOAL, Table 6 compares executed periods to GOAL (test arm minus filter off; positive = slower). The equal-actor mean weights the nine actors equally; the median, Hodges–Lehmann estimate and Wilcoxon test are pair-weighted. Conditioning on joint success selects the easier layouts and does not estimate the overall completion-time cost. Most joint-success pairs have equal times; the Wilcoxon test discards these zero differences, is pair-weighted and is computed on an outcome-selected subset, so its p-values are not inferential.

Validation versus final test (exploratory, post hoc). Table 7 sets the development-sweep estimates at $\alpha = 10$ (validation layouts, used for gain selection) beside the final-test estimates.

Stress cells (pre-specified as reported only; development data, descriptive). As the locked protocol specified, the development sweep also ran the diagnostic and stress densities $K = 4, 5, 8$ on validation layouts (Table 8). They were not part of the final test and did not affect gain selection.

Table 7: Estimates at $\alpha = 10$ on validation layouts (development data) and on the reserved final test (exploratory, post hoc comparison; 95% intervals).

K	data	pairs	fail off → on (%)	GOAL off → on (%)	GOAL loss [pp]	formation increase [m]
1	validation	1,800	5.8 → 0.0	51.1 → 47.7	3.4 [2.5, 4.3]	0.0104 [0.0085, 0.0125]
1	final test	9,000	5.4 → 0.1	54.7 → 49.9	4.8 [4.4, 5.3]	0.0117 [0.0108, 0.0126]
2	validation	1,800	17.6 → 1.6	65.1 → 53.3	11.8 [10.1, 13.5]	0.0257 [0.0230, 0.0284]
2	final test	9,000	16.6 → 0.8	66.6 → 56.8	9.8 [9.1, 10.5]	0.0202 [0.0192, 0.0212]
3	validation	1,800	46.9 → 5.6	31.9 → 22.5	9.4 [7.8, 10.9]	0.0290 [0.0262, 0.0320]
3	final test	9,000	48.0 → 3.9	32.0 → 21.5	10.6 [9.8, 11.3]	0.0278 [0.0266, 0.0290]

Table 8: Stress cells on validation layouts at $\alpha = 10$ (outcome shares in %; pre-specified as reported only; development data, descriptive; not part of the final test).

K	arm	trials	GOAL	COLLISION	INFEASIBLE	TIMEOUT
4	filter off	1,800	12.1	78.7	0.0	9.3
4	filter on (alpha = 10)	1,800	5.3	0.0	8.3	86.3
4	B-match (alpha = 10)	1,800	4.6	0.0	0.0	95.4
5	filter off	1,800	0.9	95.9	0.0	3.2
5	filter on (alpha = 10)	1,800	0.3	0.0	7.3	92.3
5	B-match (alpha = 10)	1,800	0.2	0.0	0.0	99.8
8	filter off	1,800	0.0	100.0	0.0	0.0
8	filter on (alpha = 10)	1,800	0.0	0.0	7.1	92.9
8	B-match (alpha = 10)	1,800	0.0	0.0	0.0	100.0

7 Exploratory mechanism analysis (exploratory, post hoc)

Everything in this section is exploratory and post hoc: categories, thresholds, groups and the example were chosen after the confirmatory results were known, the analyses read only saved logs, and they support no causal or confirmatory claim.

Definitions. TIMEOUTs are classified from the saved logs in this order, over the last 50 executed periods (the window): *slow progress* (flag off: the centroid-to-goal distance fell by at least 0.05 m over the window); *intervention-active nonprogress* (flag on, and the clipped nominal command violated an enabled barrier row in at least half of the window; split by obstacle and neighbour rows); *near-goal nonprogress* (flag on, not the previous class, final centroid within 0.30 m of the goal); *other nonprogress*. The names record only that the team made little terminal progress toward the goal (and, for the second, that the filter was overriding the command), not that the robots stopped or that the filter caused it. The classes are operational labels, not identified causes. "Reached goal region" uses the minimum centroid distance over the trial, the parking class the final one.

Nonprogress classes. Of 15,038 filter-on TIMEOUTs, 9,211 (61%) were intervention-active nonprogress, 9,143 of them involving obstacle rows; on the same layouts the unfiltered run collided in 5,699 cases, where sustained nonprogress replaced a collision, and reached GOAL in 2,274. Another 5,670 (38%) were near-goal nonprogress, 5,362 of them on layouts where the unfiltered run also timed out; 132 were other nonprogress and 25 slow progress.

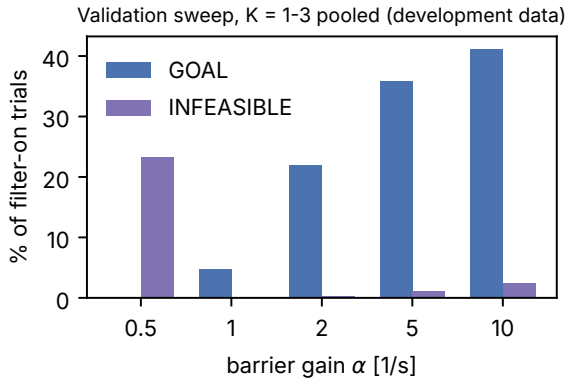


Figure 2: Validation sweep (development data): filter-on GOAL and INFEASIBLE share per gain, $K = 1$–3 pooled.

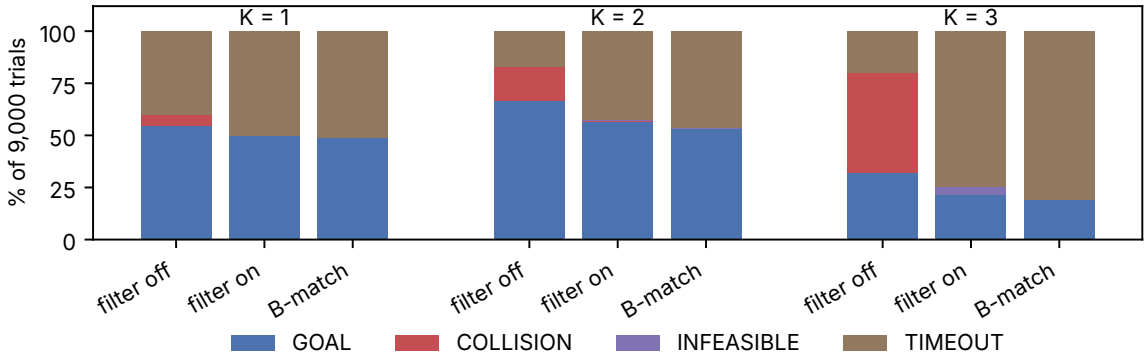


Figure 3: Final-test outcome shares per density for filter off, filter on ($\alpha^\star$) and B-match.

Lost and gained completions. Per pair, a completion is *lost* when the unfiltered run reached GOAL and the filtered run did not, and *gained* in the reverse case (Table 9, Fig. 5). Pooled over $K = 1$–3, 2,540 completions were lost and 274 gained, a net GOAL loss of 2,266 pairs; at each density the net loss divided by the number of pairs equals the criterion (b) estimate. Of the lost completions, 2,274 ended in intervention-active nonprogress, 257 in another TIMEOUT class (236 of them near-goal nonprogress), 9 as INFEASIBLE and 0 as COLLISION. The gained completions were unfiltered COLLISION (79) and TIMEOUT (195) runs; 0 came from INFEASIBLE, which the unfiltered arm never produced.

In 2,300 of the 2,540 lost pairs (90.6%; 95.1%, 89.4% and 89.8% at $K = 1, 2, 3$) the filtered run never brought its centroid within 0.30 m of the goal before any failure; in 240 (9.4%) it did. Marginal reached-goal-region rates cannot identify which pairs lost a completion; this count is paired. Entering the goal region does not by itself establish near-goal nonprogress, and the TIMEOUT classes are exploratory labels assigned from saved logs, not identified causes.

Groups A and B. Group A (2,274 pairs: unfiltered GOAL, filtered intervention-active nonprogress; 2,259 obstacle-labelled, 15 neighbour-only) and group B (11,255 pairs: both GOAL) were selected by observed outcome. The unfiltered runs of group A came closer to obstacles (median

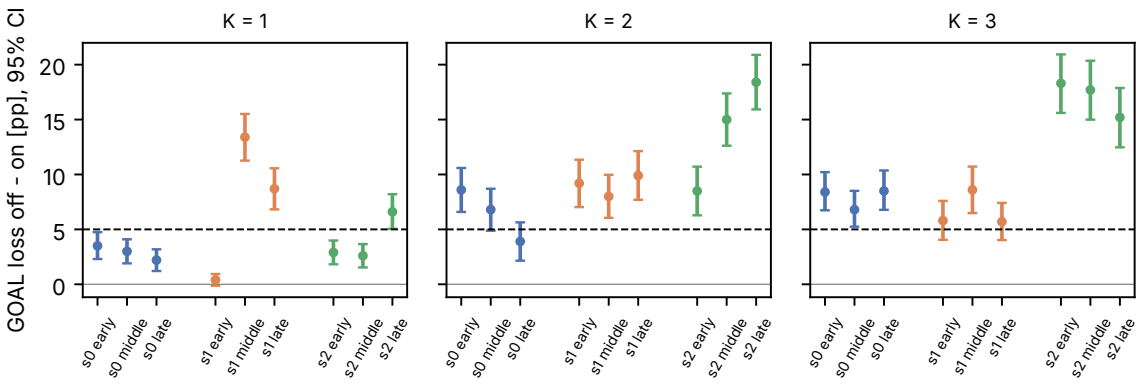


Figure 4: Per-actor GOAL loss (off minus on) with descriptive 95% Newcombe intervals, grouped by training seed; dashed line: the 5 pp margin of the pooled criterion (b).

sampled minimum surface clearance 3.3 cm, group A only) than those of group B (14.4 cm; Fig. 6). Their applied commands violated at least one filter obstacle row in 89% (A) versus 4% (B) of runs, and a robot-pair row in 85% versus 36%.

Example (exploratory, post hoc). Fig. 7 shows one group-A pair, chosen by a rule fixed before plotting: the pair whose unfiltered minimum sampled obstacle clearance is closest to the unrounded group-A median (3.30 cm), ties by actor index, density and trial. In exact decimal comparison 2 candidates were equally close; the actor-index tie-break selected actor s1_late, $K = 2$, trial 348 (clearance 3.30 cm). The unfiltered run reached GOAL after 83 periods with its closest approach at period 21; the filtered run's centroid made no further progress from period 78. At that instant, robot 0 (the robot with the smallest obstacle cover barrier, $h = 0.0046\,\mathrm{m}^2$, body clearance 9.5 cm) is shown with its cover disc. This is a geometric selection: it does not show that this robot or constraint caused the nonprogress, and sustained nonprogress of the centroid does not mean that every robot stopped.

Clearance thresholds. Of all trials, 51.1% reached GOAL unfiltered, and 43.1% did so while keeping a sampled minimum obstacle clearance of at least 5 cm (a post hoc cut equal to ℓ; Fig. 6 shows all thresholds), against a filter-on GOAL rate of 42.7% on the same layouts (Fig. 6). This describes how close the unfiltered runs came to obstacles; it does not measure what a buffer or any filter would cost. Among unfiltered GOAL trials the descriptive AUC of the unfiltered minimum obstacle clearance for "filtered run did not reach GOAL" was 0.96; it is in-sample, post hoc and partly mechanical, because obstacle rows activate at small clearance, and is not a validated predictor.

Infeasibility. All 433 INFEASIBLE trials were reproduced (433 of 433) by the unchanged filter code at the saved states in an earlier audit. An infeasible QP has no optimal active set, so the audit recorded, for each infeasible robot, the zero-command slack of every enabled row (the normalised value $(g^\top u - b)/\|g\|$ at $u = 0$, negative when the row is violated) and examined the row with the most negative slack. In every case that row's barrier value was within $4.3 \times 10^{-8}\,\mathrm{m}^2$ of its stand-off $h^\star$ (Section 3), with slack between -4.9×10^{-6} and -8.4×10^{-9}. The zero command violated between 1 and 3 enabled rows (two or more for 240 of 433 robots), and recomputing the binding slack in

Table 9: Lost and gained completions, filter off → filter on (exploratory, post hoc), in pairs out of 9,000 per density. Lost: off GOAL and on not GOAL, split by the filter-on outcome (TIMEOUT classes as in the nonprogress-class paragraph); gained: off not GOAL and on GOAL, split by the filter-off outcome. Net GOAL loss = lost − gained = off GOAL count − on GOAL count. Last rows: whether the filtered run of a lost pair reached the goal region (saved flag).

	$K = 1$	$K = 2$	$K = 3$	$K = 1$–3
Lost: intervention-active nonprogress	428	905	941	2,274
other TIMEOUT class	22	116	119	257
INFEASIBLE	0	3	6	9
COLLISION	0	0	0	0
total (% of pairs)	450 (5.0)	1,024 (11.4)	1,066 (11.8)	2,540 (9.4)
Gained: from COLLISION	4	34	41	79
from TIMEOUT	13	107	75	195
from INFEASIBLE	0	0	0	0
total (% of pairs)	17 (0.2)	141 (1.6)	116 (1.3)	274 (1.0)
Net GOAL loss off−on, pairs (pp)	433 (4.81)	883 (9.81)	950 (10.56)	2,266 (8.39)
Lost, filtered run reached goal region:				
no (% of lost)	428 (95.1)	915 (89.4)	957 (89.8)	2,300 (90.6)
yes (% of lost)	22 (4.9)	109 (10.6)	109 (10.2)	240 (9.4)

float32 changed its sign for 88 of them. 410 of 433 met a near-boundary diagnostic fixed before the (post hoc) audit computation ($|\text{slack}| \le 10^{-6}$ and $|h - h^\star| \le 10^{-3}\,\text{m}^2$). The infeasible states thus sat where stopping is only marginally inadmissible, and the evaluation of that row is sensitive to rounding. This alone does not show that the full QP would become feasible, that the zero command would satisfy every other row, or that the trial would end differently. A follow-up analysis of the same saved states (exploratory, post hoc; the filter's row construction evaluated at each state, no solver call) computed, for each of the 433 infeasible robots, the largest uniform normalised slack $t^\star$ any command in the box achieves, exactly by vertex enumeration, and all minimal conflicting row sets. In every case $t^\star < 0$ (between -6.7×10^{-7} and -5.6×10^{-10}, median -2.6×10^{-8}), and the smallest conflicting set had exactly three rows: two neighbour rows and one obstacle row in 407 cases and one neighbour and two obstacle rows in 26; the command box was not part of any minimal conflict (0 cases). The infeasible robots were thus constrained from three sides, with the three gradients positively spanning the command plane, and the resulting empty set was very thin. For 7 robots $t^\star$ was above -10^{-9}, within the solver tolerance; the enumeration solver tests only the unconstrained optimum, projections and pairwise vertices, so whether a tolerance-accepted command exists among its candidates there was not checked. The effect of these marginal conflicts on later outcomes remains unresolved. The stand-off exists at every gain, and the development sweep had INFEASIBLE outcomes at every gain, most at $\alpha = 0.5$ (1,251), so these events are not specific to $\alpha\Delta = 1$.

Transit. Writing $n = (p_i - p_j)/d$, the pair row bounds robot i's closing speed $-n^\top \dot{p}_i$ by $\alpha(h_{ij} - h^\star_{ij})/(4d)$, with $h^\star_{ij} = \delta_{ij}(d)/\alpha$. During formation transit (centroid more than 0.30 m from the goal) this cap had median 0.72–0.78 m/s, above the largest barrier-point speed $V_p \approx 0.50\,\text{m/s}$, and was binding in 1–8% of the periods in which a moving robot closed on a neighbour, so neighbour rows seldom limited transit.

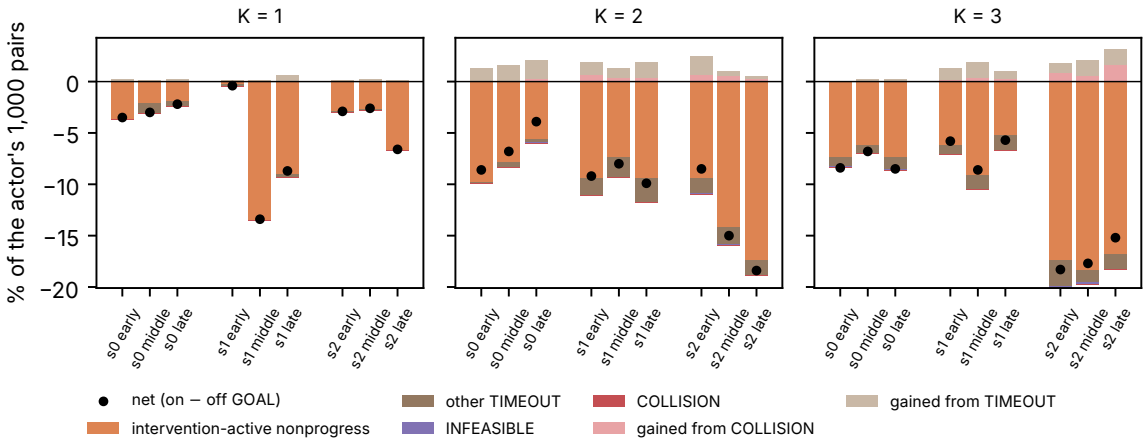


Figure 5: Lost and gained completions per actor (exploratory, post hoc), as % of each actor's 1,000 pairs, grouped by training seed. Below zero: lost completions stacked by filter-on outcome; above zero: gained completions stacked by filter-off outcome. The dot is gained minus lost, the signed change in GOAL (on minus off); Table 9 reports the net GOAL loss with the opposite sign (off minus on). No lost completion ended in COLLISION (0 pairs).

8 Discussion

Hypotheses and evidence. *H1 (collisions fall).* Descriptively consistent for these actors: with the filter on there were 0, 0 and 0 collisions against 483, 1,490 and 4,319 with the filter off. Zero filter-on collisions is consistent with the conditional guarantee of Section 3 and is numerical evidence on the evaluated trajectories; it does not certify the solver, verify every assumption or establish floating-point safety; the 6, 75 and 352 trials whose QP became infeasible ended without a fallback, so their later course is unknown, and the uncertified B-match also had no collisions. The confirmatory endpoint is the composite COLLISION-or-INFEASIBLE, whose reduction met criterion (a) at every density (Table 3); collision counts alone were not a confirmatory endpoint. *H2 (formation error and completion time rise, more so in denser layouts).* The common-prefix formation increase was 0.0117, 0.0202 and 0.0278 m at $K = 1, 2, 3$. These point estimates exceed the 0.01 m margin, and criterion (c), which tests non-inferiority, was not met at any density: non-inferiority was not established (confirmatory). This is not the same as a confirmatory test showing that the increase exceeds the margin. The estimates increase with density, but no density trend was tested, so the "more so in denser layouts" part is not established; here density changes both the obstacle count and the distance from the final training mixture ($K = 2, 5, 8$), so trends across K would not be effects of clutter alone. In pairs where both arms reached GOAL the full-trajectory formation error rose by only 0.0008, 0.0031 and 0.0043 m (pre-declared secondary, conditioned on joint success), so most of the common-prefix increase arises in pairs whose trajectories diverged. GOAL rates were lower with the filter at every density (descriptive 95% intervals for the loss exclude zero); criterion (b) tests non-inferiority and was not met, and GOAL rate is not the completion-time part of H2. In joint-success pairs the time to goal changed little (equal-actor mean 0.30, 0.68 and 2.09 periods; Table 6), and this conditional analysis does not estimate the completion-time cost. *H3 (failures under filtering come from deadlock or infeasible steps rather than slower progress).* Of the 15,471 filter-on trials that did not reach GOAL, 99.8% ended INFEASIBLE or as TIMEOUT with the protocol's sustained-nonprogress flag, and 25 were TIMEOUTs still closing on the goal at the

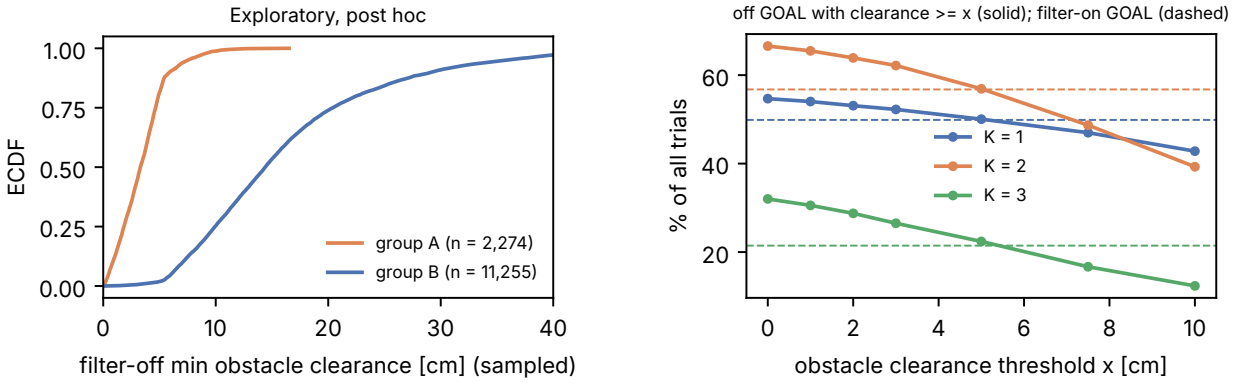


Figure 6: Exploratory, post hoc. Left: unfiltered minimum obstacle clearance (sampled), group A vs group B. Right: share of all trials that reached GOAL unfiltered with clearance at least x, per density, with the filter-on GOAL rate (dashed).

horizon (flag pre-declared as secondary, descriptive). The flag, however, accompanied 99.9, 99.9 and 100.0% of TIMEOUTs with the filter off as well (99.9, 99.8 and 99.8% with it on): at this horizon almost every TIMEOUT ends with little progress toward the goal in its last 50 periods. The flag describes that low terminal progress; it cannot distinguish a team parked near the goal from one held by obstacles or by the filter, and its high prevalence in both arms does not identify filter-specific deadlock, so it does not test H3. In the exploratory classes of Section 7, 61% of filter-on TIMEOUTs were intervention-active nonprogress and 38% near-goal nonprogress, 5,362 of the latter on layouts where the unfiltered run also timed out.

Collisions versus sustained nonprogress. With the filter on there were no collisions (0, 0 and 0 at $K = 1, 2, 3$; the remaining failures were INFEASIBLE; Table 3); in the exploratory classification most filtered timeouts were intervention-active nonprogress, mostly on layouts where the unfiltered run collided or reached GOAL. We read this as a description of where the lost completions occurred, not as a causal attribution; nonprogress of this kind may reflect the undesirable equilibria that CBF-QP filters can introduce [24], which these logs do not establish. The completions lost in group A involved unfiltered runs that passed close to obstacles (median 3.3 cm, group A only; Fig. 6), inside the 0.6–10.6 cm clearance band a stopped robot needs (Section 3), although the actors were trained with a proximity penalty below 15 cm. These lost completions are where the policy's habitual clearance and the filter's conservatism (cover disc and margin) conflict; which side to change is not identified here. The filter reached GOAL 1.0–3.2 percentage points more often than B-match on the same layouts, with more INFEASIBLE outcomes (B-match cannot be infeasible by construction).

Gain at the lemma boundary. The selected gain gives $\alpha^\star\Delta = 1$, the largest gain Lemma 1 admits. The lemma still keeps $h \geq 0$ over each period, though with no strictly positive endpoint bound from the initial barrier alone. The infeasible states sat at the stand-off, which exists at every gain, and infeasibility occurred at every gain in the development sweep, so we do not attribute it to $\alpha^\star\Delta = 1$ alone; a numerical contribution of the boundary gain is not excluded.

Predictions and an unaddressed recommendation. The locked protocol contains an analytic prediction: at $\alpha = 0.5$ the zero command becomes admissible for a robot pair only at a barrier-

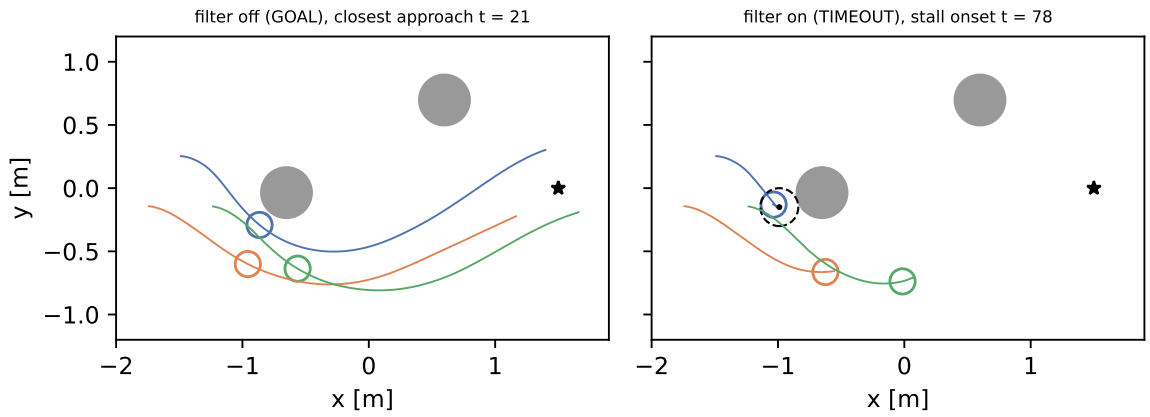


Figure 7: Exploratory, post hoc example. Selection rule (fixed before plotting): among group-A pairs, the pair whose filter-off minimum sampled obstacle clearance is closest to the unrounded group-A median; ties by actor index, density, then trial. Left: filter off, bodies at the closest-approach instant. Right: filter on, bodies at nonprogress onset (first state from which the centroid stays within 0.05 m of its final position), with the cover disc (dashed, radius $R+\ell$) of the robot with the smallest obstacle barrier at that instant (ties by robot index). Star: goal. Geometric selection only; not evidence of cause.

point separation of 0.668 m, above the nominal 0.520 m spacing, so $\alpha = 0.5$ was expected to conflict strongly with holding the formation. A methodology report written before the validation sweep, but committed only after the final test, also predicted that the selected gain would lie in $[5, 10]$. The sweep selected $\alpha^\star = 10$, the largest admissible grid value, and gave 0 filter-on GOAL outcomes with 1,251 INFEASIBLE at $\alpha = 0.5$ (Fig. 2); this concerns feasibility and completion, not a measured formation conflict. The same report estimated the implemented obstacle margin at about 13 times a tight one-sided margin at $d = D_o$, predicted that the cover-disc barrier would cost completions, and recommended advisor review of the margin; the margin was locked unchanged without that review. How much of the GOAL loss is due to margin conservatism, cover-disc geometry or the equal split was not tested.

Remark (zero-command-admissible rows). For these offset-point barriers the second-order margin can be evaluated at the held command instead of bounded over the input set: δ is replaced by a term proportional to $|\omega|$, written as two affine rows per barrier, which makes the zero command admissible whenever $h \geq 0$ in exact arithmetic. Its ingredients are existing ideas [14, 17, 18]. It would move the marginal state from the stand-off to $h = 0$ but not remove sensitivity to rounding, and it was not implemented or evaluated here.

Practical implications (suggestions, each tied to a result; not findings). (i) Pair the filter with an infeasibility fallback and a rule that resolves sustained nonprogress: with neither, 2,274 of the 2,540 lost completions ended in intervention-active nonprogress and 433 trials ended INFEASIBLE. This study does not evaluate either remedy. (ii) Treat the stand-off with an explicit tolerance rule or zero-command-admissible rows, in a separately designed follow-up (infeasibility audit). In the development sweep GOAL increased across the gain grid, while infeasibility was

non-monotonic: lowering α from 10 to 5 reduced both GOAL (2,224 to 1,936) and INFEASIBLE outcomes (129 to 61; Fig. 2). These validation results do not show that a lower gain would improve overall usefulness. (iii) In this post hoc analysis lost completions coincided with small unfiltered clearances on the same layouts (AUC 0.96, in-sample and partly mechanical); whether a clearance statistic from independent rollouts predicts a policy's filter cost was not tested. (iv) Include a brake baseline on the same rows: here B-match, which cannot be infeasible by construction and carries no safety certificate, had no collisions and reached GOAL 1.0–3.2 percentage points less often than the filter (descriptive, this scenario). (v) Where a policy habitually passes inside the clearance band a stopped robot needs, the cost may be reduced from either side (less conservative barrier geometry or margin, or training that keeps more clearance); the proximity penalty used here did not keep the group-A passes outside that band.

9 Limitations

The study uses one simulated scenario, one filter instantiation (cover disc with $\ell = 5\,\text{cm}$, two-sided state-only margins, equal split, the largest admissible gain, no fallback) and nine actors from three training runs of one custom MAPPO implementation. The actors did not meet the study's filter-off competence gate, which was pooled over $K = 2, 5, 8$; the primary densities $K = 1, 2, 3$ were chosen after their filter-off validation results had been seen; and $K = 1$ and $K = 3$ are absent from the final curriculum stage, though encountered in earlier stages. All intervals reflect layout sampling for fixed actors, not variation across training runs. The margins are the author's judgement. Validation layouts controlled curriculum advancement and were reused for gain tuning. The formation criterion covers the common executed prefix only. Time to goal is analysed only in joint-success pairs. The safety guarantee is analytic but conditional on A1–A9; A1 holds in simulation only to float32 accuracy, which the zero-collision count checks numerically. The exploratory analyses are selected by observed outcome and descriptive. Neighbour pairs were almost always within communication range, so limited communication was not tested. Three related-work classifications rest on abstracts only (Table 1).

10 Future work

Separating the causes of the lost completions needs targeted comparisons on the same actors and layouts: the tight one-sided margin against the present two-sided one, a smaller offset ℓ, zero-command-admissible rows, and an infeasibility fallback with a rule that resolves sustained non-progress. A policy family that passes the competence gate, communication ranges shorter than the formation with dropouts or delays, and sensing noise would test how far the results carry over. Any such study would use the reserved, unused 5,000,000–5,999,999 layout block under its own locked protocol and disclose which of the present post-test findings informed it.

11 Reproducibility

Code, configuration, saved logs and analyses are in `https://github.com/mahdisallam860/cbf-marl-filter-study`; its commit history records that the protocol was locked before the gain sweep and the sweep before the final test. Layout seeds: validation $2{,}000{,}000 + K$; final test $4{,}000{,}000 + 1{,}000p + K$ for actor index p; the block 5,000,000–5,999,999 is reserved and unused. The per-step logs of every final-test trial (poses, raw, clipped and applied commands, interventions, brake activations) are committed, so all analyses here can be repeated without re-running trials.

12 Protocol deviations and disclosures

- The margins and protocol decisions were chosen by the author without advisor review; advisor agreement has not been obtained.
- The protocol was locked before any filter-on data with the frozen MAPPO actors existed, but after engineering stress tests with random and adversarial stand-in controllers and demonstration runs with an obstacle-blind stand-in controller (the locked document's wording "before any filter-on data existed" is corrected here).
- Filter-off validation results had been seen before the protocol was locked.
- The checkpoint rule was committed before the checkpoints were evaluated, but after earlier validation results were seen.
- The validation layouts controlled curriculum stage advancement and were reused as development data for gain tuning.
- The primary densities $K = 1, 2, 3$ were chosen after the filter-off gate had failed and the actors' filter-off validation results had been seen.
- The methodology report was committed after the final test.
- The mechanism categories were chosen after the confirmatory results were known.
- The post-test decision rule was not committed before the mechanism results were read.
- The locked protocol labels the paired interval for criteria (a) and (b) "Newcombe method 10" but describes it as uncorrected ("without continuity correction"), which is method 8; the analysis code implements the description. With method 10 every bound changes by less than 0.001 percentage points and no criterion changes (found after the final test). The filter-on versus B-match intervals in Table 4, computed after this was found, use method 10 itself.
- One of the 27,000 final-test trials (s2_early, $K = 3$, trial 646) started with a negative obstacle cover barrier ($h = -0.00777\,\mathrm{m}^2$, body clearance 0.087 m), an exception to assumption A5; it is kept in all confirmatory numbers.
- Training seed 1 crashed at iteration 565 and was resumed from its saved iteration-546 state without restoring the environments or the global random-number state; iterations 547–565 were discarded. Actors s1_middle and s1_late come from the resumed run (Section 4).
- The near-boundary diagnostic in Section 7 was fixed before the post hoc audit computation, not in the locked protocol.

13 Conclusion

Under the pre-set criteria, this execution-time CBF-QP filter (cover disc, two-sided state-only margins, $\alpha^\star\Delta = 1$, no fallback) was not shown useful for these nine frozen MAPPO actors at any of the three densities: it reduced collision-or-infeasible outcomes but did not establish non-inferiority in goal completion or common-prefix formation error. At $K = 1$ the GOAL-loss estimate lay inside its margin and the shortfall was concentrated in one training run; at $K = 2, 3$ both estimates exceeded their margins. A brake on the same rows had no collisions and reached GOAL slightly

less often. The actors had not met the study's competence gate (pooled over $K = 2, 5, 8$), so the filter's cost on policies that keep more clearance is not known.

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